

SCOTT THOMAS ANDERSEN

Senior Software Engineer

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EXPERIENCE	Oracle - IC3 - Senior Member of the Technical Staff - Automation Tools - Improved monitoring of automation tools by proposing, designing, and developing an anomaly detection framework that now supports three major tools to identify downtime, performance issues, and validate tool integrity; greatly reducing manual review overhead and downtime by identifying over 1,200 anomalies that facilitated actionable improvements. - Designed a low-code AI workflow creation platform to facilitate easy creation and execution of LLM powered tools, ultimately leading to 22 new high impact tools across a diverse set of teams. - Cut AI token costs 72% without sacrificing tool precision by creating an AI testing framework that enables quickly testing new tool configurations and comparing costs, execution time, and scoring metrics of AI workflows. This tool was extensible to other areas and led to a collaboration excellence award.	Mexico City, Mx <i>Sep. 2024 - Present</i>
	Oracle - IC2 - Member of the Technical Staff - Automation Tools - Improved developer productivity by improving automation tools for detecting code regressions, filing bugs, and automating testing processes. - Identified a pain point in our validation process and developed a tool to automate said process preventing human error, and saving to date over 900 hours of manual labor via 3000+ job submissions. - Expanded a regression analysis tool to support multiple versions of an internal service while it migrated platforms, eliminating downtime, expanded the tool's support while adhering to strict security standards. - Refactored a major bug analysis tool from Perl to Python improving support for a tool that services issues reported by over 650 customers each day.	Mexico City, Mx <i>July 2023 - Sep. 2024</i>
	Oracle - Intern - JDBC Test Engineer - Effectively doubled the capacity of our testing tool by designing a framework to easily configure the test harness to create and execute tests in new environments. - Improved test coverage by expanding our test framework to support mocking of the database listener to make timeouts and connection failures reproducible.	Mexico City, Mx <i>Aug. 2022 - June 2023</i>
	University of Michigan - Research Assistant - C.S. Department, Dr. Kevin Leach - Decompiled hundreds of projects using industry standard binary analysis tools like Radare2, Objdump and Ghidra to create a dataset of annotated x86 assembly code matched with C/C++ code comments for a code comprehension study. - Fine-tuned BiLSTMs for assembly code summarization using Google cloud computing resources to improve comprehension of x86 assembly.	Ann Arbor, MI <i>Aug. 2019 - Aug. 2021</i> <i>Part Time</i>
	PROJECTS Cuda Enabled Ray Tracer - C++ GPU accelerated Ray Tracer using CUDA. Blizzard Detection - Expands common winter storm detection methods with ERA5 reanalysis data to improve typical winter storm detection accuracy by 5 points. Plant Tracking - YOLOv26 ML computer vision tool to track plants from video.	<u>Check them out!</u>
EDUCATION	Master of Science in Computer Science and Engineering <i>National Autonomous University of Mexico (UNAM) - GPA: 9.64/10.00</i> <i>Focus area: Signals, Images, and Virtual Environments</i>	Mexico City, Mx <i>May 2024</i>
	Weather and Climate Risk and Data Analytics Graduate Certificate <i>University of Illinois Urbana-Champaign - GPA: 3.89/4.00</i>	Remote <i>Dec. 2025</i>
	Bachelor of Science in Computer Science <i>University of Michigan Ann Arbor - GPA: 3.4/4.0</i>	Ann Arbor, MI <i>May 2021</i>
SKILLS	Programming: C++, C, Python, Bash, Java, PLSQL Scientific/ML: PyTorch, CUDA, Matlab, OpenGL, Dask Tools: Git, Google Cloud, Oracle Cloud, GDB, VIM Languages: Native English Speaker, Fluent Spanish	